Davis Baldwin

davison.baldwin@gmail.com | 630-414-4137 | Warrenville, IL 60555 | linkedin.com/in/davisonbaldwin
Product design portfolio: davisonbaldwin.github.io

Education

BHA 05/2026

Carnegie Mellon University | Pittsburgh, PA

- Phi Beta Kappa Inductee
- 3.78 GPA, University Honors, College Honors
- *Relevant Coursework*: Concepts of Mathematics, Fundamentals of Programming and Computer Science, Introduction to Data Structures, Principles of Computing, Reasoning with Data, Nature of Reason
- Vice President of The Originals A Cappella
- Languages: English (Fluent), Spanish (Fluent)

Work History

Research Fellow 05/2025 - 08/2025

Carnegie Mellon University | Pittsburgh, PA

- Conducted research in improvisational composition and reharmonization, investigating the acquisition of music as a language.
- Composed, performed, and produced a fully improvised, seven-movement piano suite as the centerpiece of the research.
- Independently performed all mixing and mastering engineering for the recorded suite.
- Listen: youtu.be/7EuehU9jKHM

Software Engineer Intern 05/2024 - 08/2024

Gamma Technologies LLC | TA Associates, Insight Partners portfolio company, Westmont IL
A physics-based engineering software solution serving automotive, energy, and data center industries

- Improved and optimized GUI components across the Gamma suite as a member of the GUI team.
- Debugged and troubleshot Java application issues across the Gamma technology stack: sophisticated Java applications, a custom distributed-computing solution, 3D graphics tooling, and high-performance data storage.
- Validated and edited technical documentation.

Projects

Universe · Interactive 3D Planetarium

- Built with Claude AI: a JavaScript/Three.js planetarium rendering 119,625 HYG-catalog stars across six connected scales: night sky, solar system, stellar neighborhood, Milky Way galaxy, cosmic web, and free-flight deep space with Sagittarius A*.
- Sky mode includes 88 IAU constellation figures, 85 classical constellation paintings, deep-sky objects, an astrophysical phenomena catalog, and NASA's exoplanet archive with 4,689 real planetary systems.
- Solar system computes accurate positions from JPL Keplerian elements: planets with NASA-imagery textures, 21 moons, satellites, named asteroids, seven comets with dynamic anti-solar tails, and trans-Neptunian dwarf planets; time simulation spans the years 1000–3000.
- Orientation-first design: guided nine-stop tour, universal search across every rendered object, data cards on click, and progressive texture loading for a fast first paint.
- Explore it live: davisonbaldwin.github.io/universe

Stock Evaluator · Index-Benchmarked Analysis

- Built with Claude AI: a Python/Flask application that scores any US stock against a total-market index fund baseline, so every verdict answers the honest question: would this beat simply buying the index?
- Analysis engine combines block-bootstrap simulations, rolling entry-point windows, and reverse discounted-cash-flow tests; compare mode benchmarks up to 10 tickers at once against a choice of 8 index funds.
- Deployed live with an interactive terminal-styled interface: stock-evaluator-mp72.onrender.com

Music Technology · Six Works

- A body of self-composed, self-produced music built inside custom visual worlds, including "Rest My Soul," a multitracked one-man choir performed and self-recorded across every vocal part, set in a custom infinite 3D environment.
- Directed the full pipeline end to end: composition, performance, recording, video direction, and post-production coordination with recording and mixing engineers.
- Watch "Rest My Soul": youtu.be/ZdAaofDbIfM

Sudoku Autosolver · Backtracking Puzzle Solver

- Built a Sudoku solver in Python using recursive backtracking, handling puzzles of varying difficulty, with a built-in hint system that surfaces the candidate numbers in each square.
- Automated detection of singleton, double, and triple candidate values to narrow the search space.
- Watch the demo: youtu.be/7QvkMWuUWXc